Gauge Absorption: Why Input-Affine FiLM Conditioning of a Mamba Selective Scan Decays to Identity on Weakly-Decodable Axes

A Small-Scale Mechanistic Null on a Single RTX 4080

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Abstract

On a single RTX 4080 Laptop GPU, we pin down a training pathology in FiLM-style regime conditioning of a Mamba selective scan. The conditioner is a per-block channel-wise affine on the block input, meant to steer the input-dependent parameters $\Delta, \mathbf{B}, \mathbf{C}$. Under joint training this modulation is a **decisive null**. It collapses to identity and resists even forced activation with direct router supervision, on the *weakly-decodable* regime axes a financial world model faces. The cause is a *gauge* symmetry. An input-side affine's scale folds exactly into the block's own $W_\Delta/W_B/W_C$ projections, and its shift, which no host bias can absorb in full, decays as an unrewarded direction, so the optimizer slides the modulation back to identity. The decay follows a dataset-invariant signature ($\tau \approx 7000$, $R^2 > 0.999$) on FI-2010 and Kaggle crypto, architecture-general by the algebra of the fold but not yet shown at FM scale. The null is separately confirmed on Polymarket, and it is *overdetermined*. An output-side placement is absorbed the same way, while a language-model control bounds the result by engaging the modulation only when the regime is strongly decodable. We establish all of this with an affordable but exhaustive apparatus, an exact bit-identical paired control, forced-active router-supervised escalations, a weight-decay ablation, a toy reference-scan, and a char-Mamba language-model control. The financial datasets are only a substrate. No backbone dominates a matched ablation, and the one surviving economic edge is architecture-independent, since a plain logistic regression reproduces it.

1. Introduction

MOSS celebrates rigorous science at small scale, and this paper is in that spirit. Every experiment runs on a **single RTX 4080**, and the $\leq$ 1-GPU constraint is what makes the study clean, not what limits it. Selective state-space models (Mamba) (Gu & Dao, 2024) have reshaped long-context sequence modeling, but *which forms of conditioning survive their training dynamics* is poorly understood. We pick one concrete, widely-attempted conditioning method, FiLM (Perez et al., 2018) regime conditioning of a Mamba backbone, and diagnose at small scale why it fails. We then turn the failure into a reusable methods package.

The object of study. We attach a *RegimeFiLM* modulator to a Mamba world-model backbone. The modulator has two parts, a router that infers a soft regime latent, and a hypernetwork that emits per-block channel-wise (γ, β). Because the selective-scan parameters $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ are input-dependent linear projections of each block's input (§2), applying the affine to the block *input* is the natural, kernel-friendly way to push a regime signal *into* the dynamics without touching the fused CUDA kernel. The modulator is zero-initialized, so at step 0 the modulated backbone is *bit-identical* to the unmodulated one.

The null, and the apparatus that establishes it. Despite that exact control, we apply the strongest activation pressure available to us, forced-active initialization *plus* direct router supervision on the most defensible regime axis. Even so, the modulation decays to identity on every objective and dataset (§3). The mechanism is what makes the null worth reporting.

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Under review for the Workshop on Methods and Opportunities at Small Scale (MOSS), COLM 2026. Do not distribute.

An input-side affine's scale is a *gauge* direction that folds losslessly into the block's own projections, so the optimizer slides it back to identity. Gauge absorption removes this constant scale on any axis. The input-dependent part of the modulation survives only when the regime is strongly decodable, so the collapse is specifically the fate of *weakly-decodable*, drop-in conditioning. The *output-side* placement the gauge argument names does not escape this either, so the null is *overdetermined*. The apparatus that establishes this is listed in the contributions below.

Contributions. (1) A **decisive, forced-active null** for input-affine FiLM conditioning of a Mamba selective scan *on the weakly-decodable regime axes we study*, with an exact zero-init control, on three datasets and a money objective (§3). (2) The **gauge-absorption mechanism** and its **dataset-invariant** signature ($\tau \approx 7000$, $R^2 > 0.999$), plus an **output-side placement showing the null is overdetermined** (App. B). (3) A **small-scale, drop-in diagnostic protocol** (§4), comprising the exact-control + escalation protocol, the toy reference-scan in-kernel demo, the weight-decay ablation, and the char-Mamba LM control bounding the null to weakly-decodable regimes.

2. Setup

Limit order books, for a non-finance reader. A *limit order book* (LOB) is the live ledger an exchange keeps for an asset. It records how much volume rests on the buy and sell sides at each price near the current mid-price, updating many times a second as orders arrive, cancel, and execute. The task throughout this paper is to forecast the next short-horizon move (price direction, volatility, or order flow) from the changing shape of this ledger. We test the same conditioning method on three structurally different books, so that one shared null across them speaks to the optimizer rather than to a single dataset's quirks. **FI-2010** is regulated equities (Nasdaq Nordic), trading on a centralized exchange with fixed hours and discrete price ticks. **Kaggle crypto** spot books trade 24/7 with no circuit breakers, giving heavier tails and thinner, faster-moving books. **Polymarket** is not a continuous price at all but a binary prediction market, so its price is a probability in $[0, 1]$ that a contract settles to 0 or 1.

A selective SSM maps an input x_t to an output y_t via a latent state $h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t$, $y_t = \mathbf{C}h_t$, where $\bar{\mathbf{A}}, \bar{\mathbf{B}}, \mathbf{C}$ are the discretized state, input, and output matrices. Mamba makes the scan parameters input-dependent:

$$\Delta_t = \sigma(W_\Delta x_t), \quad \mathbf{B}_t = W_B x_t, \quad \mathbf{C}_t = W_C x_t, \tag{1}$$

where W_Δ, W_B, W_C are learned projections, σ the softplus nonlinearity, and $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ the now input-dependent scan parameters. As a result, *any* affine transformation of x_t propagates into all three scan parameters. Mamba-3 further makes the state transition A_t a data-dependent projection of x_t, so the affine propagates there as well, which only adds to the fold. The backbone is an L=4 stack of Mamba-3 SISO (Lahoti et al., 2026) blocks. A router emits a soft regime distribution $\boldsymbol{\alpha}_t = \mathrm{softmax}(W_r x_t^{(0)}) \in \mathbb{R}^R$ from the stem output $x_t^{(0)}$, where R is the regime count (R=4 on FI-2010/Kaggle, R=8 on Polymarket). A zero-initialized hypernetwork then maps this to bounded per-block FiLM parameters $\gamma^{(\ell)} = 1 + \tanh(\tilde{\gamma}^{(\ell)})$, $\beta^{(\ell)} = \tanh(\tilde{\beta}^{(\ell)})$ (ℓ the block index, $\tilde{\gamma}, \tilde{\beta}$ the raw pre-tanh outputs), so γ=$\mathbf{1}$, β=$\mathbf{0}$ at step 0. The **input-side** site gates the block input, $\gamma^{(\ell)} \odot \mathrm{RMSNorm}(x_t^{(\ell-1)}) + \beta^{(\ell)}$, routing the regime into $\Delta/\mathbf{B}/\mathbf{C}$ via Eq. (1). A **post-scan** control gates the block output instead, sharing the identical modulator. A Switch-Transformer load-balance term (Fedus et al., 2022), disabled in the escalations, regularizes the marginal router entropy. Two diagnostics gate every claim, $\mathrm{film}_g = \mathrm{mean}\,|\gamma - 1|$ (zero iff identity) and the router entropy reg_H (uniform maximum $\log R$).

3. The Null, and Why It Holds

Joint training drives FiLM inert. Across five objectives spanning two datasets (Table 1), FiLM collapses to identity. film_g decays toward zero and reg_H sits at its uniform maximum (Table 1). Where a generalization gap reads positive (direction +0.008, settlement +0.18), FiLM is provably inert, and the treatment is *absolutely worse* in both regimes. The gap is a distributional artifact whose sign flips with head calibration, which is why the diagnostics are load-bearing.

Forced-active escalation (Polymarket). The decisive small-scale move is to fight the optimizer as hard as we can. On **Polymarket** (R=8) we force FiLM active ($\mathrm{film}_g = 0.150$ at step 0) and supervise the router *directly* on the spot efficiency-ratio axis, with the strongest settings (LR multiplier 50, supervision weight 30, decoupled router/FiLM, fed observation volatility). The modulation still monotonically decays (film_g from 0.150 to 0.043, still falling), with $\mathrm{reg}_H = \log 8$ throughout. The capacity is present but unused. On FI-2010, a router head trained alone on the *volatility* bucket reaches 0.92 agreement, so the capacity to encode that axis exists, yet even there the modulation collapses (a linear probe on the frozen

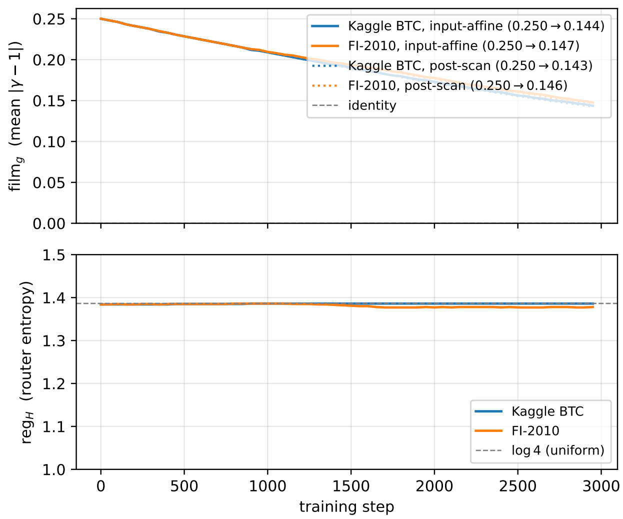

Figure 1. **Strongest-null test, and its placement-invariance.** Forcing the modulation active ($\mathrm{film}_g = 0.250$ at step 0) and supervising the router directly on the efficiency-ratio bucket still decays film_g monotonically toward identity on both datasets (top, solid = input-side), with reg_H pinned at $\log 4$ (bottom). The fit gives $R^2 > 0.999$ with a dataset-invariant $\tau \approx 7000$ and a negative fitted asymptote. The post-scan (output-side) gate (top, dotted) decays along the same trajectory, because its scale folds into the bias-free output projection just as the input-side scale folds into $W_\Delta/W_B/W_C$. The null is therefore invariant to where the affine is applied (App. B).

representation is weaker, App. B). On the *predictability* axis, the axis this escalation directly supervises, agreement is only weak (≈ 0.25–0.36), so the collapse cannot be blamed on missing regime information.

Why identity is a free direction. This instances a general *absorbability criterion*. A channel-wise affine on an activation immediately consumed by a learned linear map A has its scale absorbed into A, $A(\gamma \odot \hat{x}) = (A\,\mathrm{diag}(\gamma))\hat{x}$, with or without a bias. Its shift is absorbed only when A carries a trainable bias and no nonlinearity intervenes. Modifying the in-kernel W_Δ, W_B, W_C, by contrast, is *not* gauge-absorbable. Our input-side modulation $\tilde{x} = \gamma \odot \hat{x} + \beta$ is consumed immediately by the selection re-projections (Eq. (1)), so its scale folds into $W_{\{\Delta,B,C\}}$ with no change to the function the block computes. These projections are bias-free up to a lone dt bias that can absorb only the shift's Δ-component (App. B), so the shift as a whole cannot fold into the host; it decays instead as an unrewarded direction under reconstruction dominance (below). The scale is therefore a *gauge* direction, a flat, lossless valley the optimizer slides along. This predicts a decay rate that is a property of the parameterization, not the data. On the weakly-decodable LOB axes the supervised router collapses to uniform, so γ reduces to its constant part and weight decay folds the remainder to identity (§4).

The strongest-null escalation yields a dataset-invariant τ. We confirm that prediction with a *cross-dataset strongest-null escalation*. This time we raise the initialization further, to $\mathrm{film}_g = 0.250$, and run on two datasets with the load-balance term disabled. The modulation monotonically decays on *both* (from 0.250 to 0.144 on Kaggle, from 0.250 to 0.147 on FI-2010), with $\mathrm{reg}_H \approx \log 4$ throughout. An exponential fit $a\,e^{-t/\tau} + c$ matches both at $R^2 > 0.999$ (seed 0; the in-kernel toy below adds seeds 0–3), with near-identical parameters ($a \approx 0.30$, $\tau \approx 7000$ steps). The decay is monotone to the last logged step, with the endpoint still falling at ≈ 0.15 and identity as the extrapolated limit. The fitted asymptote also lies at or below identity ($c \lesssim -0.05$, only more negative when the lower bound on c is relaxed), so the trajectory extrapolates toward identity rather than to a positive plateau. That the rate is essentially *identical* across two very different markets points to the optimizer, not the data (Figure 1). Repeating this escalation on the *best-decoded* volatility bucket instead (which a frozen-router head fits to 0.92 on FI-2010) gives the same decay to 0.144 on both datasets ($\tau \approx 6700/6610$, $R^2 = 0.9999$, $\mathrm{reg}_H \approx \log 4$), so the collapse is gauge absorption, not an undecodable supervised axis.

The null is overdetermined. The gauge argument names an obvious next test, gating *after* the scan. Under the identical escalation, the output-side gate does *not* escape. Each block ends in a *bias-free* output projection, so the output scale folds into it, $\gamma \odot (W_{\mathrm{out}} h) = (\mathrm{diag}(\gamma) W_{\mathrm{out}}) h$ (from 0.250 to 0.146 on FI-2010, from 0.250 to 0.143 on Kaggle, same τ, $R^2 > 0.999$, seed 0 as the input-side run). The one non-absorbable component, the additive shift, decays in lockstep under reconstruction dominance (weight decay acting on a direction the reconstruction loss does not reward). The null is thus *overdetermined*. Two independent routes remove any surviving direction, and both share a single driver, weight decay over a loss that rewards neither. Gauge absorption removes every direction the host weights can represent. Reconstruction dominance removes the one direction they cannot, the non-absorbable shift. A modulation that escapes both would have to act *inside* the fused kernel, outside our hardware envelope, so we can only delineate the boundary as a location rather than a parameterization (derivations and the output-side fit in App. B).

4. Small-Scale Methods That Establish the Boundary

Three elementary probes (full numbers in App. C) turn the asserted boundary into a demonstrated one. **(i) Folding (toy reference scan).** An input-affine modulation folds into primed host projections to floating-point precision and decays to identity. The *same* modulation applied to the projected $\Delta/\mathbf{B}/\mathbf{C}$ *inside* the scan is non-absorbable instead ($\sim 10^4\times$ larger fold residual) and **persists** (seeds 0–3). It survives the very training that decays the affine form. **(ii) Gauge vs. regularization.** With weight decay off, the input-affine is no longer pulled to identity, and its decay time constant blows up $\sim 46\times$. This shows the decay is **weight-decay-driven over a flat loss**, not earned. **(iii) A language-model control bounds the null.** A 2-layer char-Mamba LM on prose vs. Python, with the domain-supervised regime and the same input-affine FiLM, does *not* collapse. With a strongly-decodable regime (router accuracy ≥ 0.9; 0.91 at the last logged step), film_g rises from 0.11 to 0.52 instead. Gauge absorption always removes the *constant* component. The input-dependent component survives only when the regime is decodable enough to be worth encoding, so the collapse is specifically the fate of *weakly-decodable*, drop-in conditioning. (Single-seed toy control; the qualitative link between a decodable router and persistence is clear, but the plateau value is not load-bearing.)

5. The Substrate Is Not the Subject

The financial datasets are a demonstration substrate. The backbone is a Mamba-3 world model in the DreamerV3/Drama lineage (Hafner et al., 2023; Wang et al., 2024), but none of this is the claim. Two checks confirm the backbone carries no weight. **(i) No backbone dominates.** A matched-parameter ablation (Mamba-1/2/3 SISO and a Mamba-3-matched pre-norm Transformer) shows no backbone leads on both FI-2010 and Kaggle. The single-seed cross-dataset gaps are within run-to-run noise (Table 3, App. A). This is exactly what a result about how selective scans *train*, rather than which one is best, should predict. **(ii) The surviving economic edge is architecture-independent.** On the identical 645-market held-out BTC set, a selective-participation gate turns a directional settlement probability into a survivable high-predictability edge (95% CI $[+\$2{,}445, +\$7{,}747]$ over the 127 traded markets). A logistic regression on the same spot-conditioned features *reproduces and exceeds* it on that same set ($+\$6{,}914$ vs. the settlement head's $+\$5{,}147$). The operative variable is therefore the features and the gate, not the sequence model (App. D).

6. Related Work

Scale redundancy and scale-invariance. That a channel-wise affine before a linear map is redundant with that map is a known theme in the normalization literature. van Laarhoven (2017), Hoffer et al. (2018), and Li & Arora (2020) show that weights feeding a normalization layer are scale-invariant, which recasts L_2/weight decay as an effective-learning-rate schedule rather than a capacity control. Our null extends this redundancy to a setting that has not been analyzed before. Here the affine is a per-regime, *input-dependent* FiLM that folds into the selective scan's own $W_\Delta/W_B/W_C$. We add an optimizer-implicating, dataset-invariant decay signature ($\tau \approx 7000$, $R^2 > 0.999$), an output-side placement that makes the null *overdetermined*, and the fact that the only kernel-friendly drop-in form is exactly the absorbable one. Mamba-2 recasts the scan as structured masked attention (Dao & Gu, 2024). DeepLOB (Zhang et al., 2019) attains 0.628 macro-F1 on FI-2010 horizon-10 labels, versus a 0.261 majority-class floor, confirming our data carries signal.

7. Discussion

Running everything on one consumer GPU makes the whole apparatus affordable to run, the exact zero-init control, the escalations, and the ablations ($\approx$2–4×10^{16} FLOPs total, App. C). The full study is $\sim$25 runs at $\sim$14 minutes each, about six GPU-hours end-to-end on one laptop GPU. It is not what makes the result valid. The conclusiveness comes from the exact-control-plus-escalation methodology, which would hold on any accelerator. The constructive controls that *bound* the null, the weight-decay-off ablation, the LM control, and the output-side fit, are single-seed (the in-kernel toy is seeds 0–3).

The null itself, by contrast, is multi-setting and multi-seed (full limitations, App. C). **Takeaway.** Do not gate a selective scan's input, or output, with a learned affine and expect it to persist. It is a gauge direction that the optimizer removes at a data-independent rate. Only placements acting on $W_\Delta/W_B/W_C$ *inside* the fused kernel survive, and that is exactly the change that breaks drop-in deployment. The toy shows such an in-kernel modulation *persists* (seeds 0–3); whether it *helps* the objective at scale is open. What transfers is the gauge algebra, which is scale-free and applies by construction wherever the affine is consumed by a learned linear map; the decay-rate empirics and the decodability boundary are demonstrated only at this scale. The practical payoff is a **drop-in diagnostic protocol** (boxed in App. C), a roughly 14-minute single-GPU test that forces the gate active and watches whether it decays, before spending GPU-months on a conditioning mechanism a matched scan will silently undo.

Use of Large Language Models

In keeping with the COLM 2026 policy on disclosure of LLM use, we report that a large language model was used while preparing this paper. It assisted in drafting and editing the prose, including the phrasing of the reference list, and was used to assist with evaluation.

A. Diagnostic tables

Family	Objective	Dataset	Gap (sign, unit)	film_g	Verdict
Volatility/scale	StudentT NLL	FI-2010	≈ 0 nats	$\approx$ ident.	NULL
Order-flow volume	Volume NLL	FI-2010	≈ 0 nats	$\approx$ ident.	NULL
Direction	macro-F1	FI-2010	$+0.008^\dagger$ F1	0.018	NULL
Settlement	BCE log-loss	Polymarket	$+0.18^\dagger$ nats	0.008	NULL
PnL	simulated trade	Polymarket	$-\$627^\dagger$	0.010	NULL

Table 1. RegimeFiLM is inert across every objective. reg_H equals its uniform maximum ($\log R$) in all rows. The Gap column is in each objective's own units, so only its sign and the inert film_g are comparable. †a non-zero gap whose sign is a distributional artifact, not a FiLM effect. The PnL (profit-and-loss) row's $-\$627$ is the FiLM A/B gap on the PnL *objective* (treatment−control), *not* an economic total (cf. the $+\$5{,}147$ settlement total, App. D). The direction row's $\mathrm{film}_g = 0.018$ is the endpoint of the *supervised* escalation on that head; the joint-trained value is ≈ 0.004 (Table 2), while the settlement (0.008) and PnL (0.010) rows are joint-trained. All are inert, but the column mixes the two measurement types.

Dataset	Asset / resolution	Gap (pred / vol)	film_g	Verdict
FI-2010	event stream	≈ 0 / ≈ 0	0.004	NULL
Kaggle BTC	1 min	≈ 0 / ≈ 0	0.004	NULL
Kaggle ETH	1 min	≈ 0	0.005	NULL
Kaggle ADA	1 min	≈ 0	0.005	NULL
Kaggle BTC	5 min	≈ 0	0.005	NULL
Kaggle BTC	1 s (slice)	≈ 0	0.005	NULL

Table 2. Cross-dataset confirmation of the regime-FiLM null ($\mathrm{reg}_H = \log 4$ in every row). The null is dataset-, asset-, and resolution-independent.

Backbone	Params	FI-2010 recon NLL	Kaggle recon NLL
Mamba-1	11.45M	$\mathbf{-0.5403}$	0.0241
Mamba-2	10.20M	-0.5309	$\mathbf{-0.0213}$
Mamba-3 SISO	10.30M	-0.5346	0.0207
Transformer (pre-norm)	9.61M	-0.5041	0.0224

Table 3. Matched-parameter backbone ablation (held-out reconstruction NLL, lower is better). No backbone wins both datasets. Mamba-1 leads on FI-2010, Mamba-2 on Kaggle. Mamba-1 is 11.45M, $\approx 11\%$ over the matched ~ 10.3M budget inherited from the parent configuration. Excluding it, the FI-2010 ordering is unchanged (Mamba-3 SISO -0.5346 still ahead of Mamba-2 -0.5309). The under-budget Mamba-3-matched pre-norm Transformer is also competitive but non-dominant, so the non-dominance is not an artifact of an under-tuned or over-budget baseline.

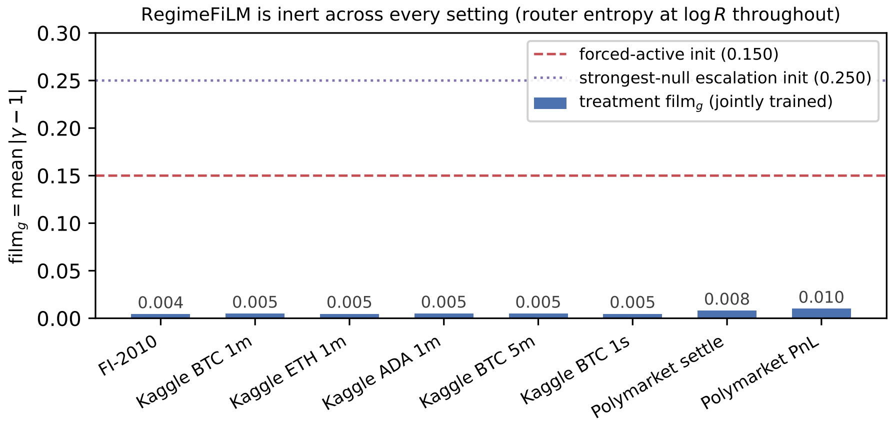

Figure 2. RegimeFiLM is inert across every A/B setting. The jointly-trained film_g (bars) sits near 0.004–0.010 everywhere, while forced-active (0.150) and strongest-null (0.250) initializations are dashed. The modulation never grows from its exact identity initialization.

B. Mechanism: the output-side placement in full

The output-side gate applies $y' = \gamma \odot y + \beta$ to the block output. Each Mamba-3 block ends in a *bias-free* linear output projection $y = W_{\text{out}} h$, so a channel-wise output scale folds exactly into it, $\gamma \odot (W_{\text{out}} h) = (\mathrm{diag}(\gamma) W_{\text{out}})h$, the same gauge move as the input side but on the block's other face. The additive shift β is genuinely non-absorbable. There is no output bias, and a nonlinear $\mathrm{RMSNorm}$ separates it from the next block's projections. Yet under forced activation and ER supervision, film_g and its additive counterpart $\mathrm{film}_b = \mathrm{mean}\,|\beta|$ monotonically decay in lockstep on both datasets (film_g from 0.250 to 0.146 on FI-2010, from 0.250 to 0.143 on Kaggle; film_b tracks within 0.005), with $\mathrm{reg}_H \approx \log 4$ throughout. The fit gives the same dataset-invariant signature ($a \approx 0.30$, $\tau \approx$ 7065 FI-2010 / 6598 Kaggle, asymptote $c \lesssim -0.05$ and below identity with the bound relaxed, $R^2 > 0.999$). Two probes localize the failure. The jointly-supervised router is uniform per step ($\approx \log 4$), with near-chance ER agreement (0.25–0.31). A frozen linear probe on the unmodulated features decodes the predictability bucket only weakly (0.33–0.36 vs. 0.25) and the volatility bucket somewhat better (0.43–0.47); this measures linear decodability of the representation, distinct from the 0.92 a supervised router head reaches on the volatility bucket on FI-2010 (§3). Yet the volatility-axis gap is also null.

Input-side shift decomposition. The input-side shift decomposes along the block's four consumption paths. The Δ-path ends in a lone trainable dt bias inside the softplus (suppressed in the simplified Eq. (1)), so the Δ-component of a constant shift folds into it exactly; this is the only host parameter that can absorb any part of the shift. The $\mathbf{B}/\mathbf{C}$ components are blocked by per-state norms interposed between the projections and the kernel (a post-norm bias cannot absorb a pre-norm offset), and the gate and scan-input paths carry no bias at all. The shift as a whole therefore cannot fold into the host, and it decays as an unrewarded direction under reconstruction dominance, exactly as observed.

C. Methods detail

Drop-in diagnostic protocol. (1) Force the gate active (e.g. $\mathrm{film}_g \approx 0.15$–$0.25$ at step 0) and supervise the router directly on the candidate regime axis. (2) Train ~3,000 steps (~14 min on one consumer GPU). (3) Fit $a\,e^{-t/\tau} + c$ to the film_g trajectory. Decay toward identity with $c \lesssim 0$ means the placement is gauge-absorbable and a matched scan will silently undo it; a rise to a positive plateau means the conditioning is being used.

All toy experiments use the pure-PyTorch reference scan that `mamba_ssm` ships (the slow path the fused-kernel hardware limit does not touch), in seconds on the same single 4080. **Folding.** The recovered projections satisfy $W_\bullet^{\text{fit}} \approx W_\bullet^{\text{base}}\,\mathrm{diag}(\gamma)$ to round-off (input-affine residual 1.2×10^{-6}). The in-kernel placement, by contrast, leaves a residual 1.3×10^{-2}, because a per-channel scale on the post-softplus Δ cannot be matched by any linear reparameterization of the pre-softplus projection. **Dynamics.** Input-affine decays from 0.086 to 0.010 with weight decay on; in-kernel persists, rising from 0.093 to 0.175 (seeds 0–3). **Weight-decay ablation.** With weight decay off, the input-affine stays near its forced-active init (toy, from 0.086 to 0.342; full-model τ goes from ≈ 6685, R^2=0.9999, to $\approx 3.1 \times 10^5$, R^2=0.48, a $\sim$46× blow-up). **LM control.** A 2-layer char-level Mamba LM, built on `mamba_ssm` blocks, trained on prose vs. Python with a context-aware domain-supervised router and the same input-affine FiLM. The router reaches ≥ 0.9 accuracy (0.98 at peak, 0.91 at the last logged step), and film_g rises from 0.11 to 0.52. We run Mamba-3 SISO throughout because the MIMO kernel's dynamic shared-memory footprint exceeds the 4080 cap at every warp-valid (rank, chunk) configuration. This is the same kernel region a surviving modulation would have to occupy.

Estimated compute (FLOPs). Each pretraining run is $\sim$3000 optimizer steps at a 64×64-token batch on FI-2010/Kaggle (128×64 on Polymarket), giving $T \approx 1.2$–2.5×10^7 tokens per run. With $N \approx 10.3$M backbone parameters, the dominant cost is $6NT \approx 0.8$–1.5×10^{15} FLOPs per run ($\sim$14 min on the 4080). The full study, the null table, the cross-dataset sweep, the input- and output-side escalations, the weight-decay ablation, and the five-backbone matched ablation ($\sim$25 runs total), costs $\approx$2–4×10^{16} FLOPs. The toy reference-scan and char-LM controls are negligible by comparison. Every experiment is thus $\gtrsim 10^3\times$ below the 10^{20} frontier-scale reference, on one consumer GPU.

Limitations. The *null* is multi-setting and multi-seed; the *constructive* claims that bound it are smaller. The optimizer attribution rests on a τ matched across two datasets, plus the within-model weight-decay ablation. The full-model weight-decay-off ablation and the LM control are single-seed, though the in-kernel toy adds seeds 0–3. The in-kernel "survives" result is at toy reference-scan scale, since the MIMO placement is outside the 4080 envelope. The substrate is finance-only, so the architecture-universality of the gauge algebra is argued, not demonstrated at language-model scale. None of these bears on the null itself.

D. Deflated economic coda

Given the spot path that defines Polymarket settlement, the model produces an over-confident but directional settlement probability. We trade it with a selective-participation strategy that bets only when a train-frozen efficiency-ratio gate fires. All totals below are on the *identical* 645-market held-out BTC set, against a mandatory no-model naive control that loses $-\$1{,}008$. The counts 127 and 243 are the per-arm traded/evaluable subsets a market-block bootstrap resamples, not different evaluation sets.

On that set the gated settlement head totals $+\$5{,}147$, a 3-seed expected-value head averages $+\$4{,}779$, and a logistic regression on the same spot-conditioned features *reproduces and exceeds* both at $+\$6{,}914$ (Brier score 0.146, the mean-squared error between forecast probability and outcome, lower is better). Each is a survivable, naive-beating edge. The settlement head's high-predictability block has $\mathrm{P(profit)} \approx 1.0$ with 95% CI $[+\$2{,}445, +\$7{,}747]$ over its 127 traded markets, and the logistic reaches $\mathrm{P(profit)} = 0.9994$ with 95% CI $[+\$2{,}844, +\$10{,}880]$ over its 243. The operative variable is therefore the features and the gate, not the sequence model. The settlement head's low-predictability block is marginal ($+\$51$, $\mathrm{P} = 0.63$); a direct expected-value head makes it robust instead ($\mathrm{P} = 0.98$–0.99), collapsing the high-vs-low PnL spread $6.2\times$ at comparable total. Only the predictability gate is confirmatory; the depth-band and early-market cuts are exploratory, and the reliability figures are recorded-vintage structural diagnostics.

References

T. Dao and A. Gu. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In *Proc. ICML*, 2024.

W. Fedus, B. Zoph, and N. Shazeer. Switch Transformers: Scaling to trillion parameter models with simple and efficient sparsity. *JMLR*, 23:1–39, 2022.

A. Gu and T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *Proc. COLM*, 2024.

D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.

E. Hoffer, R. Banner, I. Golan, and D. Soudry. Norm matters: Efficient and accurate normalization schemes in deep networks. In *Proc. NeurIPS*, 2018 (arXiv:1803.01814).

A. Lahoti, K. Y. Li, B. Chen, C. Wang, A. Bick, J. Z. Kolter, T. Dao, and A. Gu. Mamba-3: Improved sequence modeling using state space principles. In *Proc. ICLR*, 2026 (arXiv:2603.15569).

Z. Li and S. Arora. An exponential learning rate schedule for deep learning. In *Proc. ICLR*, 2020 (arXiv:1910.07454).

E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proc. AAAI*, 2018.

T. van Laarhoven. L2 regularization versus batch and weight normalization. *arXiv preprint arXiv:1706.05350*, 2017.

W. Wang et al. Drama: Mamba-enabled model-based reinforcement learning is sample and parameter efficient. In *Proc. ICLR*, 2025 (arXiv:2410.08893, 2024).

Z. Zhang, S. Zohren, and S. Roberts. DeepLOB: Deep convolutional neural networks for limit order books. *IEEE Transactions on Signal Processing*, 67(11):3001–3012, 2019.